

House Price Prediction – Project Summary

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices using property characteristics such as area, bedrooms, bathrooms, stories, parking facilities, furnishing status, and other amenities. The project aimed to identify the factors influencing house prices and evaluate different machine learning models for accurate price estimation.

Data Preparation and Analysis

The Housing dataset was loaded and explored using Python libraries such as Pandas and NumPy. Data preprocessing included checking for missing values, removing duplicate records, and converting categorical variables into numerical form using One-Hot Encoding. Exploratory Data Analysis (EDA) was performed to understand the dataset structure, feature relationships, and price distribution.

Model Development

The dataset was divided into training (80%) and testing (20%) sets. Two machine learning models were trained and evaluated:

- Linear Regression
- Random Forest Regressor

Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score.

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,021,546	1,400,566	0.612

Linear Regression achieved the best performance and was selected as the final prediction model.

Key Findings and Insights

- House area was identified as one of the strongest factors influencing property prices.
- Properties with more bathrooms, additional stories, parking facilities, and better furnishing status generally had higher prices.

- Correlation analysis confirmed that property features have a significant impact on market value.
- The Actual vs Predicted visualization showed that the Linear Regression model produced reasonably accurate predictions.
- Despite being a more advanced algorithm, Random Forest did not outperform Linear Regression on this dataset, indicating that the relationship between features and price is largely linear.

Business Recommendations

Real estate companies can use the developed model to estimate property prices more efficiently and support pricing decisions. Buyers and sellers can also benefit from understanding how features such as area, parking availability, furnishing quality, and amenities affect property value. Focusing on these factors can improve property valuation accuracy and market competitiveness.

Conclusion

This project successfully demonstrated the application of machine learning in house price prediction. Linear Regression was selected as the best-performing model with an R^2 Score of 0.653, explaining approximately 65.3% of the variation in house prices. The project highlights the importance of data-driven decision-making in the real estate sector and provides a practical foundation for future predictive analytics applications.

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Project Title: House Price Prediction

Tools Used: Python, Jupyter Notebook, Pandas, Scikit-learn, Matplotlib, Seaborn